

4(a)	any four from: 1 (random / spontaneous) mutation gave resistance ; 2 pathogen / disease / plague, was selection pressure ; 3 <u>mutation</u>, gave selective advantage / was selected for / allowed crayfish to survive (plague) ; 4 survivors / resistant crayfish, bred / reproduced / passed on mutation ; 5 frequency of, <u>allele</u> / <u>mutation</u>, (for resistance) increased ; 6 natural / directional, selection ; 7 AVP ;	4
4(b)(i)	1 (heat to) 90–98 °C so DNA, denatures / strands separate ; 2 (cool to) 50–65 °C so primers, bind / base pair / anneal (to DNA) ; 3 (heat to) 68–75 °C so, <i>Taq</i> / DNA, polymerase, makes DNA / (new) strand ; 4 repeat cycles to amplify DNA ;	4
4(b)(ii)	used, a (named) database / bioinformatics ;	1

Question	Answer	Marks
4(c)(i)	A ;	1
4(c)(ii)	$(2^3) = 8 ;$	1
4(d)(i)	(named) plasmid / vector (used) ; (containing) gene / DNA, for (green) fluorescent <u>protein</u> ; (containing) a (prokaryotic) promoter ;	3
4(d)(ii)	to act as a, marker / reporter, <u>gene</u> or to identify, recombinant / GM / transgenic, bacteria / organisms or to show expression of another transferred gene ;	1

Question	Answer	Marks
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